Arduino-Based Shadow Alarm

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Shadow alarms are usually used for protection against theft. A shadow alarm is a device that sounds an alarm when a shadow falls on it. Described here is a simple circuit of an Arduino-based shadow alarm. This compact shadow alarm unit is capable of sensing a moving shadow in a restricted area, and can be easily installed on a wall, window or door to protect your valuables from theft. Constant lighting is required in the confined area to detect the moving shadow.

Circuit and working

The circuit of Arduino-based shadow alarm is shown in Fig. 1. It consists of an Arduino board, power supply, light-dependent resistor (LDR) sensor, buzzer, relay driver and a few other components. Arduino Uno board is the heart of this circuit.

Arduino. The Arduino is a popular microcontroller platform that has taken the world by storm. It is an Open Source physical computing platform based on a simple yet powerful microcontroller board and development environment to write software for the board.

The latest Arduino Uno (refer Fig. 2) is a microcontroller board based on ATmega328. It has 14-digital input/output pins (of which six can be used as PWM outputs), six analogue inputs, a 16MHz crystal oscillator, a USB connection, a power jack, an ICSP header and a reset button. Just connect it to a computer through a standard USB

Test Points

Test point	Details
TP0	0V, ground
TP1	9V power supply
TP2	5V power supply
TP3	High when shadow is detected
TP4	High when shadow is detected
TP5	High when shadow is detected (LED1 glows)
TP6	High when shadow is detected

PARTS LIST

Semiconductors:
T1, T2, T3 - BC547 npn transistor
D1, D2, D3 - 1N4007 rectifier diode
LED1 - 5mm light-emitting diode

Resistors (all ¼-watt, ±5% carbon):
R1, R4, R5 - 1-kilo-ohm
R2, R3 - 470-ohm
VR1 - 4.7-kilo-ohm preset
LDR1 - Light-dependent resistor

Capacitors:
C1 - 470µF, 25V electrolytic
C2 - 100µF, 16V electrolytic

Miscellaneous:
S1 - On/off toggle switch
RL1 - 9V, 1C/O relay
PZ1 - Piezobuzzer
BATT.1 - 9V PP3 battery
- 9V DC power adaptor
CON1 - Seven-pin connector
- Arduino UNO board
CON2 - Two-pin connector

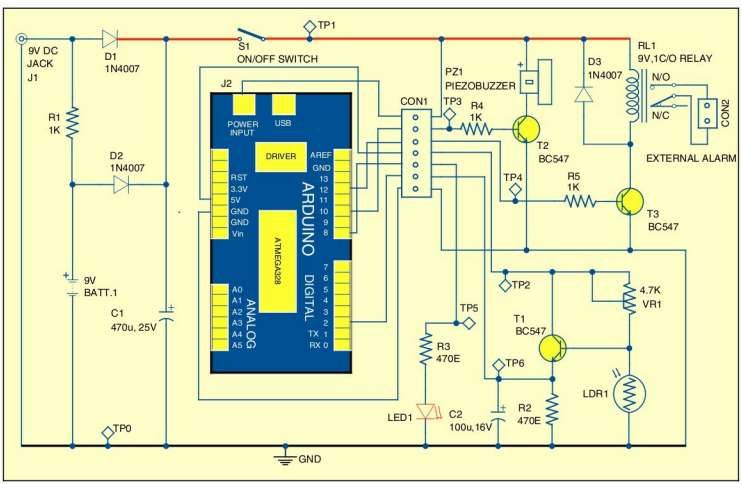


Fig. 1: Circuit of Arduino-based shadow alarm